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**SPACE TESTING OF ELECTROMAGNETICALLY SENSITIVE MATERIALS FOR BREAKTHROUGH
PROPULSION PHYSICS**

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Recently it has been experimentally demonstrated that non-local (instantaneous) communication between two beams of light (spin direction) can randomly occur. The effect is described as a consequence of the physics of quantum mechanics. Other research has experimentally demonstrated that the zero point radiation/fields (ZPF), which pervade space-time, can effectively be shielded so as to force two parallel plates together (Casimir effect). The NASA Marshall Space Flight Center is investigating the possibility that a spinning superconductor can cause a reduction in the weight of nearby objects. Recent ultrahigh-intensity, peta/eta-watt tabletop lasers have achieved extreme electric and magnetic fields, pressure, temperature and space-time curvature that can only be found close to a black hole horizon. Based on these research and experimental efforts, some examples of space experiment and materials/technology testing approaches have been examined to determine the feasibility and potential benefits of using the International Space Station (ISS) to address breakthrough propulsion physics and technology challenges. The ISS's microgravity environment, combined with the access to an extreme vacuum and plasma environment, offers some unique advantages for the testing of various electromagnetically "sensitive" materials, and associated physical interactions. The use of this space laboratory could enable much more rapid progress in the identification and optimization of anomalous and highly nonlinear effects. A modular Breakthrough Propulsion Physics testbed could be developed using one or two mid-deck locker equivalent volumes in a pressurized Express Rack, which would enable the operation or testing of various experiments and devices as part of a long duration, breakthrough physics and technology testing program. Similarly, an Express Pallet adapter site could be used as a breakthrough propulsion physics testbed for those experiments or technology demonstrations which require direct access to the space environment and larger operating volumes. Breakthrough propulsion physics experiments could also be integrated into (or use extra space in or on) micro-/nano- technology (MNT) experiments which are already under development. The International Space Station's research capabilities provide important opportunities for achieving early breakthroughs in physics understanding, and the associated materials/technology interactions, needed to accelerate advances in space vehicle/platform maneuvering and transport.

Introduction

A number of recent advances in theoretical and experimental physics and materials development are providing insight into exploratory experiment approaches, which can be used to predict and look

* NASA (and its Breakthrough Propulsion Physics Program) supports the use of International Space Station's payload capabilities for breakthrough propulsion physics research described in this paper. NASA does not, however, endorse this paper."

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for anomalies which could lead to substantial progress towards NASA's Breakthrough Propulsion Physics objectives. The International Space Station (ISS), which will soon begin to conduct a wide range of research activities, provides a unique environment for these type of exploratory investigations. In late 2000 and early 2001, the ISS will have the capability to install and operate pressurized payloads or experiments in the U.S. Laboratory.

The microgravity environment, access to an extremely low particle density (vacuum), and plasma characteristics of near-Earth orbit provide a unique laboratory testing capability. Using the pressurized Express Rack, mid-deck locker equivalent (MLE) volumes (8 MLEs per Express Rack), experiments can be quickly and easily installed, operated and returned